Possible Problems for Quiz 8 to be taken 3/21/19. Of these problems I will choose 3 to be on the quiz.

Problem 1

Can the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ be diagonalized? Justify your answer.

Problem 2

If a matrix A has eigen values $\lambda_1 = 1$, $\lambda_2 = 2$ with corresponding eigen vectors $\vec{u}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{u}_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ what is A?

Problem 3

Show that the set of all second-degree polynomials is a vector space.

Problem 4

Show that for the vector space $C(R)$ (continuous functions on R) the subset of functions $f \in C(R)$ such that $f(4) = 0$ is a subspace of $C(R)$

Problem 5

Can a finite set of vectors span P, the set of all polynomials with real coefficients?

Problem 6

Determine if the set $\{\sin^2(x), \cos^2(x), 2\}$ is linearly independent in $C[0, \pi]$

Problem 7

Determine if the set $\{\sin(x)\,, \sin(2x)\,, \sin(3x)\}$ is linearly independent in $C[0, \pi]$